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Short communication

Natural Language Processing as a tool to evaluate emotions in conservation conflicts

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ABSTRACT

Conservation conflicts involve an important emotional component that has been little investigated so far. In particular, the study of emotions involved in conservation conflicts have been limited in their scope (e.g. negative vs. positive sentiments) by the language studied (mostly English). Natural Language Processing (NLP) approaches combined with informed expert knowledge constitute a promising tool to analyze the occurrence of discrete emotions in textual contents. We applied NLP on the textual content of German print and online news publications featuring the return of wolves (*Canis lupus*) in the Saxony region of Germany (n = 3096), to investigate the occurrence of 7 basic emotions and their associations with 9 stakeholder groups. All emotions could be detected with different accuracy levels, and except for *interest* that was detected in all articles, negative emotions tended to dominate the articles' emotional content (*anger* in 74% and *fear* in 36% of the articles). *Anger* was most frequently detected in articles featuring *farmers* (20%) and *hunters* (15%), as was *fear* (17% with *farmers*, 14% with *hunters*). Our study of a diverse set of emotions, in local language, demonstrates the usefulness of NLP approaches to broaden the understanding of local conservation conflicts, in which the news organizations have a pivotal, yet often neglected role.

1. Introduction

Conservation conflicts and the underlying tensions between stakeholders over wildlife management can generate heated debates (Chapron and López-Bao, 2020; Nyhus, 2016), involving strong emotions. The role of emotions in conservation conflicts has been recognized in recent years (Chapron et al., 2014; Jacobs and Vaske, 2019; Jacobs, 2012; Nelson et al., 2016), as they influence our dispositions towards wildlife (Jacobs and Vaske, 2019) and seem tied to attitudes (Landon et al., 2019; Sponarski et al., 2015; Straka et al., 2020). However, they remain seldom studied. To date, studies on emotions towards wildlife have mostly focused on fear (Flykt et al., 2013; Johansson et al., 2012). The current lack of knowledge on emotions in the context of conservation probably owes to the lack of consensus around the definition of emotions (Izard, 2007; Jacobs, 2012), their diversity (Jackson et al., 2019) and their cultural and language specificities (Aman and Szpakowicz, 2007). The study of emotions in the context of conservation conflicts thus requires strong interdisciplinary integrations and appropriate tools to address these challenges.

In recent years, Natural Language Processing (NLP) approaches, i.e. Artificial Intelligence techniques that allow computers to read and interpret information from natural language texts (Thessen et al., 2012), have been broadly used in multidisciplinary biodiversity projects (Thessen et al., 2012). Recent studies have highlighted the usefulness of Artificial Intelligence methods in conservation, as they can be used to identify conservation icons (Ladle et al., 2016), track illegal wildlife trade on social media (Di Minin et al., 2018; Xu et al., 2019), investigate human-nature interactions (Toivonen et al., 2019) or sentiments towards iconic species like white rhinoceros (Fink et al., 2020). We argue that NLP approaches, combined with informed expert knowledge (Dadvar et al., 2014; Hirschberg and Manning, 2015; Hovy, 2015; Mohammad, 2016) constitute a promising avenue to investigate emotions in conservation conflicts, as they can be applied to a broad set of textual resources, to a wide range of contexts, and most importantly, to

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datasets containing an amount of data that cannot be processed manually. However, to date, automatic detection of emotions in textual content (i.e. sentiment analysis) applied in conservation have been limited by (i) the dichotomy between positive and negative emotions, although emotions are more diverse than simple polarizations (see Hausmann et al., 2020 for a recent illustration in South African National Parks), and (ii) the focus on English language resources, which is, in many cases, of limited local conservation relevance.

In this study, we applied NLP in automatic analysis of emotions on the textual content of print and online news publications featuring the return of wolves (*Canis lupus*) in the Saxony region of Germany, where they have been present since their recolonization of Germany in 2000 (Reinhardt et al., 2019). Wolf conservation is one of the most contentious conservation topics in human-dominated landscapes (Darimont et al., 2018), and its return to Germany has sparked important media coverage. Stories involving returning carnivores are good candidates for being selected by news organizations, as they are perceived to be relevant to a broad readership (although few people interact with carnivores), involve a charismatic animal and often feature negative news (e.g. depredation events, concerns for human health) (Harcup and O'Neill, 2001). Hence, human-carnivore coexistence is frequently featured in the press (Bombieri et al., 2018; Fernández-Gil et al., 2016; Killion et al., 2018) and contains an emotional component that plays an important role in the mechanisms behind the effect of news frames on opinions, or framing effects (Lecheler et al., 2015). Here, we investigated a set of seven basic emotions, namely *interest*, *surprise*, *joy*, *sadness*, *fear*, *anger* and *disgust* (Jacobs et al., 2014). In addition, we investigated the co-occurrence of the seven emotions and nine specific wolf conservation stakeholders, namely *politics*, *research*, *Non-Governmental Organizations*, *farmers*, *hunters*, *military*, *news agencies*, *media*, *Saxony's Wolf Information Office*.

2. Materials and methods

2.1. Corpus of news

Since wolf return to Saxony in 2000, the Wolf Information Office "Kontaktbüro Wölfe in Sachsen" has been continuously compiling press clippings featuring wolves, in print and digitized formats. The sampling protocol has evolved through time: at first, articles were collected manually by surveying local (i.e., from the region of Saxony) and national press, and opportunistically collecting news on wolves from other federal states; then, the advent of online newspapers allowed automatized data collection based on keyword entries. It is therefore possible that some articles were missed, but we assume that this would have mostly affected articles from sources outside Saxony, and we do not anticipate specific biases in terms of topics or emotional content. We obtained 4000 newspaper articles, all in German, that covered the entire period from 2000 to 2018, to investigate their content. As most articles were indeed published in Saxony ($n = 2556$) or were from national sources ($n = 764$), we excluded articles from German federal States other than Saxony and focused on a set of 3320 newspaper articles. Each article was scanned and processed with an Optical Character Recognition program. We only extracted the textual information and excluded articles containing unrecognized text ($n = 224$), resulting in a total of 3096 articles for our analysis.

2.2. Natural Language Processing

We pre-processed the texts using the Natural Language Toolkit in Python: we removed stop words and non-informative symbols and punctuations, and used stemming to reduce them to their word stem. The identification of emotions followed several steps (see Supporting Information Fig. S1). First, we developed our own lexicons to identify the seven emotions in the articles' texts, by compiling words referring to each of the seven emotions. We used the SentimentWortschatz (V1.8) dictionary (Remus et al., 2010) to further enhance our list and find synonyms to the words associated with each of the seven emotions (see Supporting Information Table S1). Second, we used the Term Frequency-Inverse Document Frequency (TF-IDF) numerical approach (Aizawa, 2003) to identify the most informative and distinct words in each text and to investigate whether they included the words related to the emotions. Third, we identified the presence of emotions in each text based on the frequency of occurrence of words related to a specific emotion (Whitelaw et al., 2005). Fourth, we used a word-level n-gram model (here bigram, $n = 2$) that takes into account the co-occurring words within a given window around the words associated with the specific emotions. This provided the possibility to further look into the context that a word has been used for, rather than only the frequency of its occurrences.

2.3. Stakeholder categorization and extraction

We created a lexicon for each of the stakeholder groups by populating a list of keywords based on our own expertise. We further improved the lexicon using an inductive approach and adding terms that we found in the newspapers during the labelling process (see Evaluation). We included *politics*, *research*, *Non-Governmental Organizations*, *farmers*, *hunters* as these groups are most active in wolf conservation debates (König et al., 2020). We further included *military* (military training areas have been critical for wolf recolonization in Germany) (Reinhardt et al., 2019), *news agencies*, *media* (i.e. people involved in cultural and artistic activities associated with wolves e.g. books, films, exhibits, etc.) and the *Wolf Information Office* as these were frequently mentioned in the articles. We then used the lexicons as a proxy to detect the occurrence of the stakeholders in each article (see Supporting Information Table S2, Fig. S1).

2.4. Evaluation

The evaluation dataset consisted of 498 articles manually and independently labelled for presence/absence of the seven emotions by two annotators with a social-ecological background. The inter-annotator agreement was 0.80 (average agreement over the 7 emotions) meaning that in 80% of the labelled cases, the reviewers labelled the newspaper articles to the same emotion categories (see Table 1).

We used the manually labelled dataset, where both annotators were in agreement, to evaluate the accuracy of the NLP approach and to report the precision and the recall of the identified emotions. For a specific emotion, precision is the fraction of accurately detected instances (true positives) among the detected instances by the NLP (true positives + false positives), whereas recall is the fraction of accurately detected instances (true positives) among the total amount of relevant instances, i.e. texts that genuinely contained that specific emotion (true positives + false negatives). Accuracy also reported the F1-Score which is the number of correctly identified emotions divided by the number of all correctly identified emotions returned. Finally, to investigate the co-occurrence of emotions and stakeholder groups in the newspaper articles, we used pairwise Pearson correlation tests between the seven emotions and the nine stakeholder groups, and implemented a bonferroni correction on *p*-values to account for multiple tests.

3. Results

The classification accuracy was different among emotions (Table 1). *Disgust* (1.00) and *interest* (0.84) had the highest precision whereas *joy* (0.43) and *surprise* (0.62) had the lowest. *Interest* (1.00) and *anger* (0.61) had the highest rates of recall whereas *disgust* (0.08) and *joy* (0.09) had the lowest rates. Hence, *interest* (0.91) and *anger* (0.63) had the highest F1-scores, whereas *disgust* (0.14) and *joy* (0.15) had the lowest (Table 1). Across 3096 newspaper articles on wolves, *interest* (1.00), *anger* (0.74) and *fear* (0.36) were the most frequently detected emotions (Fig. 1).

Table 1
Annotators' categorization (among $N_{total} = 498$), inter-annotator agreement and classification accuracy (precision, recall and F1-scores) of the automatically identified emotions across 498 newspaper articles from Saxony, Germany.

	Annotator 1 categorization (N articles)	Annotator 2 categorization (N articles)	Inter-annotator agreement	Precision	Recall	F1-score
Interest	402	269	0.72	0.84	1.00	0.91
Anger	275	235	0.86	0.65	0.61	0.63
Fear	220	162	0.83	0.76	0.41	0.53
Sadness	189	143	0.80	0.68	0.34	0.46
Surprise	209	100	0.72	0.62	0.13	0.21
Joy	185	150	0.85	0.43	0.09	0.15
Disgust	147	67	0.81	1.00	0.08	0.14

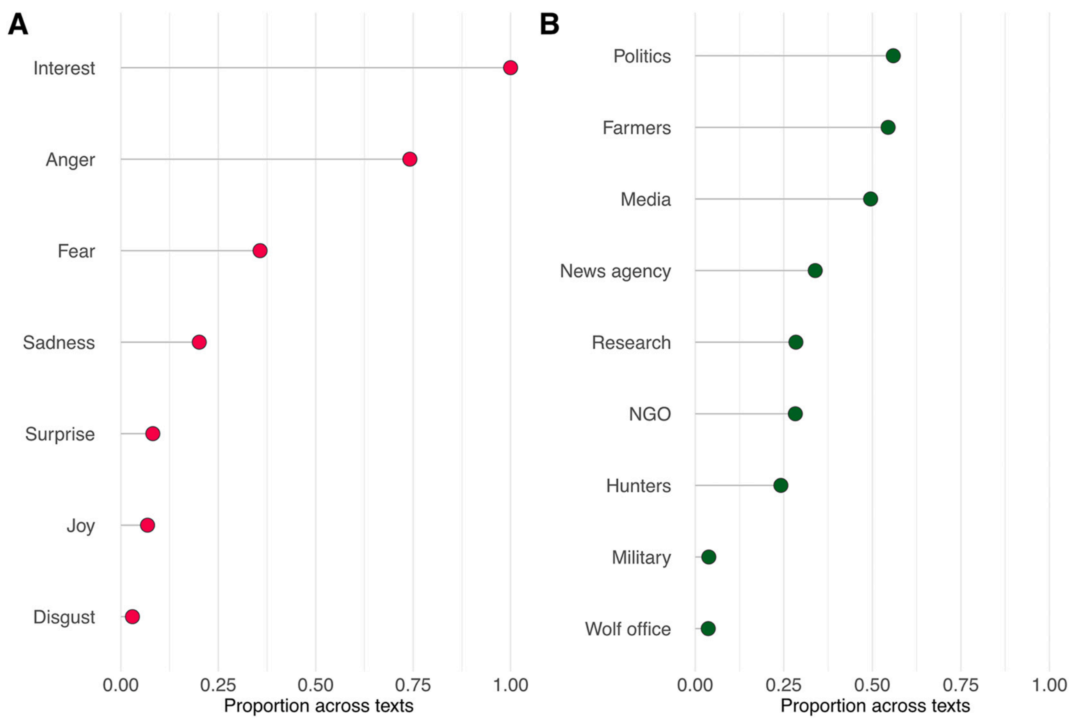

Fig. 1. Distribution of emotions (A) and stakeholders (B) across 3096 press articles reporting on the wolf topic in Saxony, Germany.

We identified *politics* (0.56) and *farmers* (0.54) as the most frequently found stakeholders in the articles (Fig. 1). Comparatively, the *Wolf Information Office* (0.04) and the *military* (0.04) were the least featured stakeholders in the articles (Fig. 1). The most frequently detected co-occurrences between emotions and stakeholders in the newspaper articles were between *farmers* and *anger* (0.20, $t = 11.33$, $p < 0.001$), *farmers* and *fear* (0.17, $t = 9.68$, $p < 0.001$), *hunters* and *anger* (0.15, $t = 9.68$, $p < 0.001$) and *hunters* and *fear* (0.14, $t = 7.82$, $p < 0.001$) (Fig. 2). The highest correlation between two emotions was found between *anger* and *fear* (0.36, $t = 21.57$, $p < 0.001$).

4. Discussion

4.1. Main findings

Overall, our approach using NLP to detect emotions in German articles featuring wolves demonstrated that all emotions could be detected in the corpus of 3096 articles, and that the approach could be applied to another language than English, increasing its local relevance for conservation. Although emotion detection accuracy was sometimes low, we detected a domination of negative emotions like *anger* and *fear*. The detection of *fear* and *anger* in news articles featuring *farmers* and *hunters*, although reflecting the challenges of human-carnivore coexistence, may hint at an unbalanced representation of wolf conservation in the news.

4.2. Natural Language Processing' relative efficiency to detect emotions

Emotion detection is a challenging task because it is language- and topic-specific (Aman and Szpakowicz, 2007). The inter-annotator agreement of ca. 80% illustrated the challenge of labelling texts containing specific emotions. However, this value was in the range of previous sentiment analysis in German (Momtazi, 2012) and other languages (Mozetič et al., 2016). Furthermore, the F1-scores, reflecting quantity and quality of detected emotions, were also in line with previous findings (Balahur et al., 2012; Kušen et al., 2017), reporting higher precision for negative emotions and lower recall values for positive emotions (Table 1), meaning that negative emotions are more accurately detected, and positive emotions occur less frequently in the news. Such pattern probably echoes the news tendencies to select and frame negative news more frequently (Harcup and O'Neill, 2001). *Interest* may not be a relevant emotion as it was detected in all newspaper articles, which publication probably already reflects an expression of interest, thereby artificially inflating recall values (1.00).

Except for *interest*, the prevalence of negative emotions like *anger*, *fear* and *sadness* reflected the overall negative framing of human-carnivore coexistence in the news (Bombieri et al., 2018; Nanni et al., 2020). The domination of *anger* and *fear* is problematic for carnivore conservation, as emotions play an important role in mediating news framing effects on people's attitudes (Lecheler et al., 2015). In fact, it is

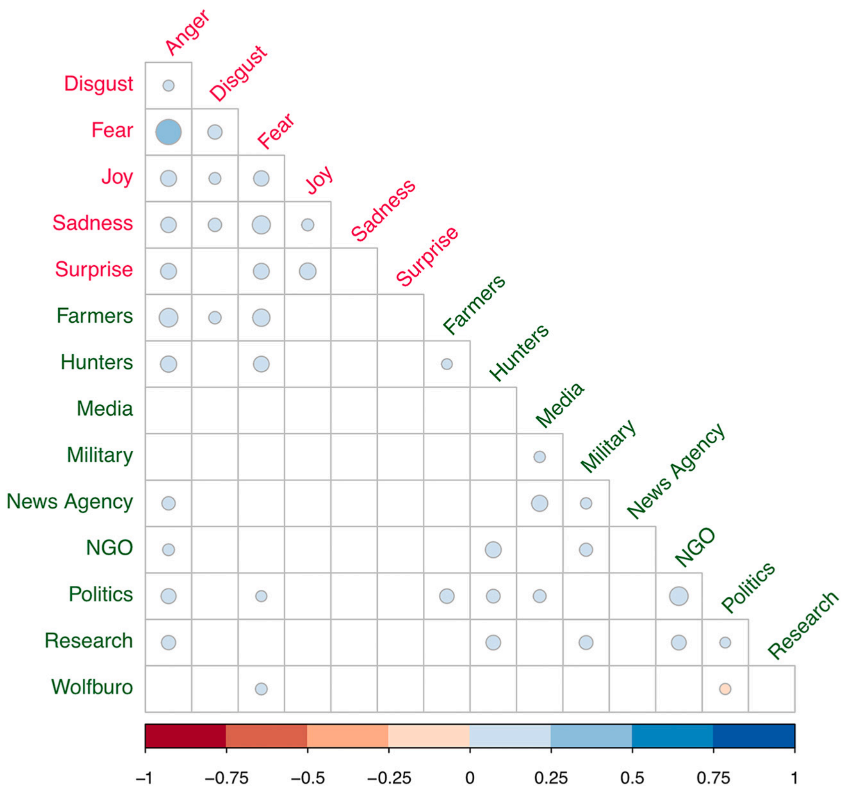

Fig. 2. Correlations between emotions (pink) and stakeholders (green) across 3096 press articles reporting on the wolf topic in Saxony, Germany. Non-significant correlations (5% threshold, *p*-values adjusted with bonferroni correction) are blanked. We excluded *interest* from the plot, as it was detected in all newspaper articles (recall = 1.00, see Fig. 1). The size of the circles represents the correlation strength. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

widely accepted that negative information has a stronger impact on people's attitudes than positive information (Soroka, 2006). As only few people directly interact with wolves (Arbieu et al., 2020) or other conflict-prone wildlife, human-wildlife relations are mainly shaped by information and not personal experience. Therefore, at a time where human activities increasingly encroach on wildlife habitats and potentially generate more conservation conflicts, the conservation community should embrace the complexity of emotions in conservation conflicts, for long-term human-wildlife coexistence. We demonstrated that NLP approaches are promising tools in this regard.

4.3. *Emotions and stakeholders' representation*

News organizations play an important role in giving a voice to conservation actors, and stakeholder recognition is a critical aspect of conservation conflict management (Redpath et al., 2017). We are aware of only few studies investigating stakeholder occurrence in the media (see e.g. Rust, 2015; Webb and Raffaelli, 2008). Our approach using NLP to investigate the co-occurrence between stakeholder and emotions revealed that the highest associations between *hunters*, *farmers*, and *anger* and *fear* depicted a bleak image of human-wolf coexistence in Saxony. This finding is at odds with a previous study that found rather neutral attitudes expressed by the broad public towards wolves in this region (Arbieu et al., 2019), reflecting an attenuation of the conflict over time. This discrepancy between the negative and emotional content found in the newspapers that focus on the short-term, often negative news and people's rather neutral attitudes that reflect longer timeframes of learning to coexist with carnivores, illustrates the news agencies' narrow focus and perception of human-carnivore coexistence. Understanding stakeholder's as well as news agencies' perceptions (Bennett, 2016) may be critical to restore a more balanced and fair representation of human-carnivore coexistence in the media. In addition, the prevalence of *politics* and the under-representation of *research* demonstrates that the information conveyed by news organizations may not always be evidence-based, and calls for improved communication from the scientific community. Therefore, NLP approaches can be a useful tool to investigate local conservation conflicts and potential polarizations (Webb and Raffaelli, 2008). Besides, they are useful to analyze big textual datasets and resources, such as online social media platform contents and news archives. However, most existing tools and technical resources, such as training datasets and gold standards for automatic detections have mainly been developed for the English language. Therefore, having these tools and classification models developed for other languages, such as our study in German, could be beneficial to local conservation contexts. In particular, this will help identifying imbalances in stakeholder representation, investigating emotional contents, and developing strategies to produce evidence-based information that do not mainly focus on short-term negative news.

4.4. *Outlook*

This work calls for further investigations of the role of emotions in news featuring conservation topics. First, to improve the accuracy of emotion detection, we would recommend manually labeling an evaluation dataset that covers sufficient numbers of each emotion to train a machine learning model, prepared by at least 3 annotators. Second, further investigations could go beyond emotion occurrence and address emotional intensity, and investigate changes at different spatial and temporal scales. Such investigations would be very important in contexts of rapid social-ecological changes. Third, comparisons across charismatic species could inform on discrepancies in species framings (Bombieri et al., 2018). Most importantly, we encourage conservation scientists to embrace interdisciplinarity and develop collaborations with scholars across journalism studies, natural and computer sciences.

CRediT authorship contribution statement

UA, AN and TM conceptualized the manuscript. UA, AN and MD developed the methodology. MD developed the Natural Language Processing approach. KH collected the data from the press articles. UA, AN and KH performed the analyses. UA prepared the results visualization. UA prepared the initial draft of the manuscript and all authors contributed substantially to the redaction of the final version.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biocon.2021.109030.

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